# Introduction

Einstein metrics : Riemannian manifolds $(M, g)$ with

$$Ric(g) = \lambda g \qquad \lambda \in \mathbb{R}.$$

Hard to find "interesting" examples

e.g. non-flat compact $(M, g)$ with $Ric(g) = 0$ ?

Kähler geometry adds more structure, leads to many examples.

e.g. Zero set $z_0^{n+1} + \dots + z_n^{n+1} = 0$ in $\mathbb{CP}^n$ has Ricci flat metric. (Yau)

# Course outline

## Part I Yau's theorem.

In Kähler geometry Einstein eq$^n$ can be reduced to scalar PDE for $f : M \longrightarrow \mathbb{R}$ :

$$\det\Big(\overbrace{h_{i\bar{j}} + \frac{\partial^2 f}{\partial z^i \partial \bar{z}^j}}^{g_{i\bar{j}}}\Big) = e^{F - \lambda f} \det(h_{i\bar{j}}) \qquad (*)$$

equiv. to $Ric(g) = \lambda g$

where $h, F$ are given, $\lambda \in \mathbb{R}$.

<u>Main PDE thm</u> (Yau) If $\lambda = 0, -1$ then a solution $f$ of $(*)$ satisfies apriori estimate

$$\|f\|_{C^3} \leq C(h, F).$$

$\rightsquigarrow$ Yau's existence thm in cases $Ric(g) = -g$, $Ric(g) = 0$

<u>Key point</u>: Sequence of solutions $f_i$ to $(*)$ cannot blow up if corresponding $h_i, F_i$ are controlled.

## Part II

How to take limits of sequences $(M_i, g_i)$ of Riemannian manifolds?

Two extremes:

(A) Fixed manifold $M$ with sequence $g_i$ such that

$$g_i \to g_\infty \quad \text{in } C^k \text{ sense to metric } g_\infty$$

(B) Sequence $(M_i, g_i)$ converges to metric space $(Z, d)$ in Gromov–Hausdorff sense

### Two general theorems

**Thm 1** (Cheeger, Gromov) If for a $C > 0$

$$|Rm(g_i)| < C, \quad \mathrm{diam}(g_i) < C, \quad \mathrm{vol}(g_i) > C^{-1}$$

then a subsequence converges in sense (A) (with $k=1$)

**Thm 2** (Gromov) If $Ric(g_i) > -C g_i$, $\mathrm{diam}(g_i) < C$ then a subsequence converges in sense (B)

## Part III Cheeger-Colding theory

<u>Informal statement</u> If $Ric(g_i) > -Cg_i$,

$diam(g_i) < C$, $vol(g_i) > C^{-1}$

then subsequence converges in sense (A) $(k<1)$ away from codimension 2 set.

## Part IV Kähler case

<u>Thm</u> (Donaldson-Sun, Liu-Sz.)

Let $(M, g_i)$ be Kähler, polarized ($g_i$ are induced by line bundles)

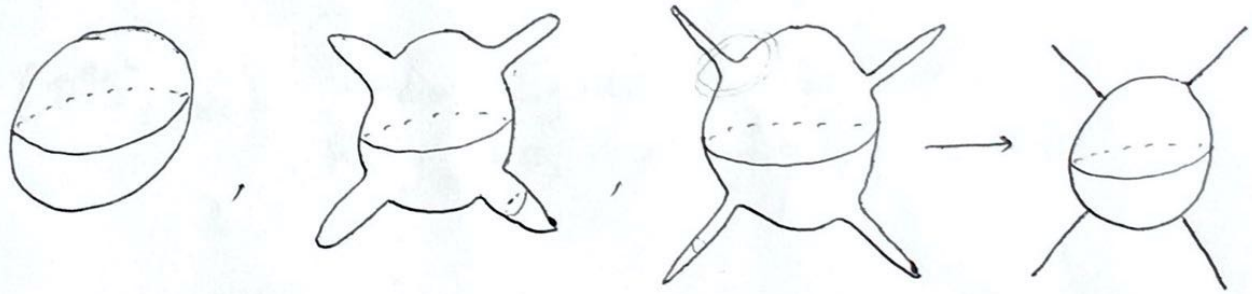
$Ric(g_i) > -Cg_i$, $diam(g_i) < C$, $vol(g_i) > C^{-1}$.

Subsequence converges to $(Z, d)$, homeomorphic to complex projective variety.

Leads to existence of Kähler-Einstein manifolds with $Ric(g) = g$ (Chen-Donaldson-Sun)

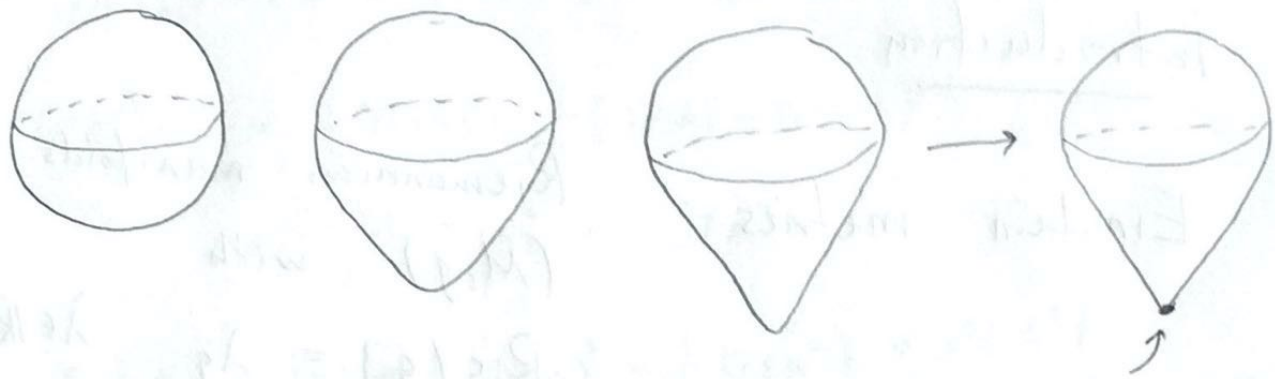
circle

thinner and thinner tori



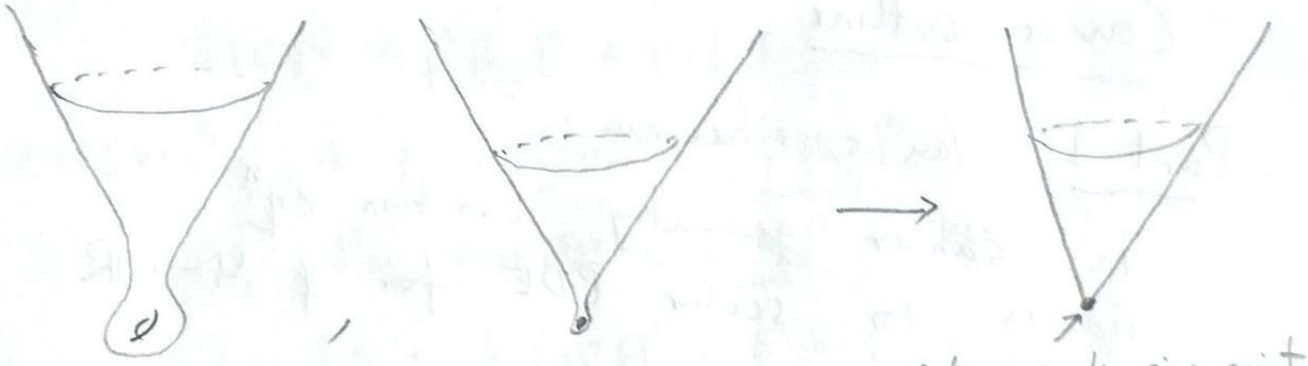
Spheres with bumps

sphere with spikes



Spheres developing cone point  Singular point

$(T^*S^2, g_{EH})$  4-dim Eguchi-Hanson metric
$Ric(g_{EH}) = 0$, asymptotically conical



$g_{EH}$  $2^{-i} g_{EH}$  Codimension 4 sing. point
flat $\mathbb{C}^2/\mathbb{Z}_2$

# Kähler Geometry

**Definition** A complex manifold $M$ is a smooth manifold with a cover $M = \bigcup_\alpha U_\alpha$ together with homeomorphisms $\varphi_\alpha : U_\alpha \to V_\alpha \subset \mathbb{C}^n$ such that transition maps

$\varphi_\alpha \circ \varphi_\beta^{-1}$ are holomorphic on their domains.

$\varphi_\alpha$ defines local holomorphic coordinates on $U_\alpha$

**E.g.** Complex projective space $\mathbb{CP}^n$ is quotient space $(\mathbb{C}^{n+1} \setminus \{0\}) / \sim$ where $(z_0, \dots, z_n) \sim (\lambda z_0, \dots, \lambda z_n)$ for $\lambda \in \mathbb{C}$

denote equiv. class by $[z_0 : \dots : z_n]$

$z_0, \dots, z_n$ are homogeneous coordinates

$$\mathbb{CP}^n = \bigcup_{i=0}^{n} U_i$$

$$U_i = \{ [z_0 : \dots : z_n] \mid z_i \neq 0 \}$$

$$\varphi_i : U_i \longrightarrow \mathbb{C}^n$$

$$[z_0 : \dots : z_n] \longmapsto \left( \frac{z_0}{z_i}, \dots, \overset{\text{omit}}{\widehat{\frac{z_i}{z_i}}}, \dots \frac{z_n}{z_i} \right)$$

$\mathbb{CP}^n$ is compact.

<u>E.g.</u> Projective manifolds

If $f_1, \dots, f_k$ are homogeneous polynomials in $z_0, \dots, z_n$, then their common zero set

$$V = \left\{ [z_0 : \dots : z_n] \;\middle|\; \begin{array}{l} f_i(z_0, \dots, z_n) = 0 \\ \text{for } i = 1, \dots, k \end{array} \right\} \subset \mathbb{CP}^n$$

is well defined.

If $V$ is smooth then it is a complex manifold.

<u>Definition</u> An <u>almost complex structure</u> on a smooth manifold $M$ is

$$J : TM \longrightarrow TM$$

such that $J^2 = -\mathrm{Id}$.

E.g. On a complex manifold if $z^1, \dots, z^n$ are local holomorphic coords and $z^i = x^i + \sqrt{-1}\, y^i$, let

$$J\left(\frac{\partial}{\partial x^i}\right) = \frac{\partial}{\partial y^i}, \qquad J\left(\frac{\partial}{\partial y^i}\right) = -\frac{\partial}{\partial x^i}$$

transition maps are holomorphic $\Longrightarrow$ $J$ is well defined

Complexified tangent and cotangent bundles

$$T^{\mathbb{C}}M = TM \otimes_{\mathbb{R}} \mathbb{C} \qquad\qquad \Omega^1_{\mathbb{C}} M = \Omega^1 M \otimes_{\mathbb{R}} \mathbb{C}$$

local bases $\left\{ \frac{\partial}{\partial z^1}, \dots, \frac{\partial}{\partial z^n}, \frac{\partial}{\partial \bar{z}^1}, \dots, \frac{\partial}{\partial \bar{z}^n} \right\}$ $\qquad \left\{ dz^1, \dots, dz^n, d\bar{z}^1, \dots, d\bar{z}^n \right\}$

$$\frac{\partial}{\partial z^i} = \frac{1}{2}\left(\frac{\partial}{\partial x^i} - \sqrt{-1}\frac{\partial}{\partial y^i}\right), \quad \frac{\partial}{\partial \bar{z}^i} = \frac{1}{2}\left(\frac{\partial}{\partial x^i} + \sqrt{-1}\frac{\partial}{\partial y^i}\right) \qquad \begin{aligned} dz^i &= dx^i + \sqrt{-1}\, dy^i \\ d\bar{z}^i &= dx^i - \sqrt{-1}\, dy^i \end{aligned}$$

$$T^{\mathbb{C}}M = T^{1,0}M \oplus T^{0,1}M \qquad \Omega^1_{\mathbb{C}} = \Omega^{1,0}M \oplus \Omega^{0,1}M$$

$\text{span}\left\{\frac{\partial}{\partial z^i}\right\}$ $\qquad$ $\text{span}\left\{\frac{\partial}{\partial \bar{z}^i}\right\}$

$\sqrt{-1}$ eigenspace of $J$ $\qquad$ $-\sqrt{-1}$ eigenspace of $J$

higher degree forms $\quad \Omega^r_{\mathbb{C}}M = \bigoplus_{p+q=r} \Omega^{p,q}M$

$(p,q)$-forms $\Omega^{p,q}M$ spanned by

$$dz^{i_1} \wedge \cdots \wedge dz^{i_p} \wedge d\bar{z}^{j_1} \wedge \cdots \wedge d\bar{z}^{j_q}$$

exterior derivative decomposes as $d = \partial + \bar{\partial}$

$$\partial : \Omega^{p,q}M \longrightarrow \Omega^{p+1,q}M$$
$$\bar{\partial} : \Omega^{p,q}M \longrightarrow \Omega^{p,q+1}M$$

$$d^2 = 0 \iff \partial^2 = 0, \ \bar{\partial}^2 = 0$$
$$\partial\bar{\partial} + \bar{\partial}\partial = 0$$

<u>E.g.</u> 1. $f : M \to \mathbb{C}$ is holomorphic

if and only if $\bar{\partial} f = 0$

2. For $f : M \to \mathbb{R}$

$\sqrt{-1}\, \partial\bar{\partial} f$ is a real $(1,1)$-form

If $f : \mathbb{C} \to \mathbb{R}$,

$$\sqrt{-1}\, \partial\bar{\partial} f = \frac{1}{2}\left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}\right) dx \wedge dy$$

# Hermitian and Kähler metrics

$M$ complex manifold with complex structure $J$.

<u>Definition</u> 1. A Riemannian metric $g$ on $M$ is <u>Hermitian</u> if $g(JX, JY) = g(X, Y)$ for all $X, Y$.

2. If $g$ is Hermitian its associated 2-form is

$$\omega(X, Y) = g(JX, Y)$$

$\omega$ is a real $(1,1)$-form.

$g$ is <u>Kähler</u> if $d\omega = 0$.

In local coordinates a Hermitian $g$ is determined by components

$$g_{j\bar{k}} = g\left(\frac{\partial}{\partial z^j}, \frac{\partial}{\partial \bar{z}^k}\right)$$

Since $\quad g\left(\frac{\partial}{\partial z^j}, \frac{\partial}{\partial z^k}\right) = g\left(\frac{\partial}{\partial \bar{z}^j}, \frac{\partial}{\partial \bar{z}^k}\right) = 0$

$g_{j\bar{k}}$ defines a positive definite Hermitian matrix at each point:

$$\overline{g_{j\bar{k}}} = g_{k\bar{j}}$$

and if $(\xi_1, \dots, \xi_n) \in \mathbb{C}^n \setminus \{0\}$ then

$$\sum_{j,k} g_{j\bar{k}} \xi_j \bar{\xi}_k > 0 .$$

Associated 2-form is

$$\omega = \sqrt{-1} \sum_{j,k} g_{j\bar{k}} \, dz^j \wedge d\bar{z}^k$$

$g$ is Kähler if $\frac{\partial}{\partial z^i} g_{j\bar{k}} = \frac{\partial}{\partial z^j} g_{i\bar{k}}$

<u>E.g.</u> (Fubini-Study metric)

Recall projection map $\pi : \mathbb{C}^{n+1} \setminus \{0\} \longrightarrow \mathbb{CP}^n$

If $U \subset \mathbb{CP}^n$ is open we define a <u>section over $U$</u> to be a holomorphic

$$s : U \longrightarrow \mathbb{C}^{n+1} \setminus \{0\}$$

such that $\pi \circ s = id$

Given a section $s$ define

$$\omega_{FS} = \sqrt{-1}\,\partial\bar\partial \log \|s\|^2$$

This is well defined on $\mathbb{CP}^n$:

if $s'$ is another section over $V$

then $s' = fs$ on $U \cap V$ for $f : U \cap V \to \mathbb{C} \setminus \{0\}$ holomorphic

so
$$\begin{aligned}\sqrt{-1}\,\partial\bar\partial \log \|s'\|^2 &= \sqrt{-1}\,\partial\bar\partial \log |f|^2 \|s\|^2 \\ &= \sqrt{-1}\,\partial\bar\partial \log \|s\|^2 + \sqrt{-1}\,\partial\bar\partial \log |f|^2 \\ &= \sqrt{-1}\,\partial\bar\partial \log \|s\|^2 .\end{aligned}$$

To check that the components of $\omega_{FS}$ define positive definite matrices

note that $\omega_{FS}$ is $U(n+1)$ invariant

$\Longrightarrow$ enough to work at one point

at $[1 : 0 : \dots : 0]$ use local coordinates

$$w^i = \frac{z_i}{z_0} \quad \text{for } i = 1, \dots, n$$

and section $s(w^1, \dots, w^n) = (1, w^1, \dots, w^n)$ over $U_0$

$$\omega_{FS} = \sqrt{-1}\,\partial\bar\partial \log\left(1 + |w^1|^2 + \dots + |w^n|^2\right)$$

at origin $\omega_{FS} = \sqrt{-1} \sum_{i=1}^{n} dw^i \wedge d\bar{w}^i$

E.g. If $V \subset \mathbb{CP}^n$ projective manifold then $\omega_{FS}$ restricted to $V$ defines a Kähler metric.

A Kähler form $\omega$ on $M$ defines deRham cohomology class $[\omega] \in H^2(M, \mathbb{R})$

<u>Theorem</u> ($\partial\bar\partial$-lemma) Let $M$ be a compact Kähler manifold. Let $\omega, \eta$ be real $(1,1)$-forms such that $[\omega] = [\eta]$. Then $\exists f : M \to \mathbb{R}$ such that

$$\eta = \omega + \sqrt{-1}\,\partial\bar\partial f$$

<u>Theorem</u> (normal coordinates) If $g$ is a Kähler metric then around any $p \in M$ $\exists$ holo. coordinates $z^1, \dots, z^n$ such that the components $g_{i\bar j}$ satisfy

$$g_{i\bar j}(p) = \delta_{ij} \quad , \quad \frac{\partial}{\partial z^k} g_{i\bar j}(p) = \frac{\partial}{\partial \bar z^k} g_{i\bar j}(p) = 0$$

# Covariant derivatives and curvature

Let $(M, g)$ be Kähler
$\omega$ associated 2-form (Kähler form)
$\nabla$ Levi-Civita connection so $\nabla g = 0$

In holo. normal coords $J$ is constant so $\nabla J = 0$
$\omega(X,Y) = g(JX, Y)$ so $\nabla \omega = 0$.

In holo. coordinates $z^1, \dots z^n$ we let

$$\nabla_i = \nabla_{\frac{\partial}{\partial z^i}}, \quad \nabla_{\bar{i}} = \nabla_{\frac{\partial}{\partial \bar{z}^i}}, \quad \partial_i = \frac{\partial}{\partial z^i}, \quad \partial_{\bar{i}} = \overline{\partial_i} = \frac{\partial}{\partial \bar{z}_i}$$

<u>Note</u>: • $J(\nabla_j \partial_k) = \nabla_j J \partial_k = \sqrt{-1}\, \nabla_j \partial_k$

so $\nabla_j \partial_k \in T^{1,0}$.

Define Christoffel symbols $\Gamma^i_{jk}$ by

$$\nabla_j \partial_k = \sum_{i=1}^n \Gamma^i_{jk} \partial_i$$

• $\nabla_{\bar{j}} \partial_k \in T^{1,0}$ and $\nabla_k \partial_{\bar{j}} \in T^{0,1}$

but $\nabla$ is torsion free, so

$$\nabla_{\bar{j}} \partial_k - \nabla_k \partial_{\bar{j}} = [\partial_{\bar{j}}, \partial_k] = 0$$

$$\Longrightarrow \quad \nabla_{\bar{j}} \partial_k = \nabla_k \partial_{\bar{j}} = 0.$$

- $\nabla_{\bar{i}} T = \overline{\nabla_i \bar{T}}$ for any tensor

In sum $\nabla$ is determined by the $\Gamma^i_{jk}$

and $\Gamma^{\bar{i}}_{\bar{j}\bar{k}} = \overline{\Gamma^i_{jk}}$. Torsion free : $\Gamma^i_{jk} = \Gamma^i_{kj}$

<u>E.g.</u> 1. $$\nabla_i dz^k = -\sum_{j=1}^{n} \Gamma^k_{ij} dz^j$$

2. Let us write a tensor such as

$$\sum_{j,\bar{k}} a_{j\bar{k}} dz^j \otimes d\bar{z}^k$$

as simply $a_{j\bar{k}}$.

Then

$$\nabla_{\bar{p}} a_{i\bar{j}} = \partial_{\bar{p}} a_{i\bar{j}} - \Gamma^{\bar{l}}_{\bar{p}\bar{j}} a_{i\bar{l}}$$ (sum over repeated indices)

Applied to metric tensor $g_{i\bar{j}}$, since $\nabla g = 0$

we have

$$\Gamma^{\bar{l}}_{\bar{p}\bar{j}} g_{i\bar{l}} = \partial_{\bar{p}} g_{i\bar{j}}$$

so using inverse $g^{i\bar{l}}$ we get

$$g^{i\bar{k}} \partial_{\bar{p}} g_{i\bar{j}} = \Gamma^{\bar{l}}_{\bar{p}\bar{j}} g_{i\bar{l}} g^{i\bar{k}} = \Gamma^{\bar{l}}_{\bar{p}\bar{j}} \delta^{\bar{k}}_{\bar{l}} = \Gamma^{\bar{k}}_{\bar{p}\bar{j}}$$

conjugate : $\Gamma^i_{jk} = g^{i\bar{l}} \partial_j g_{k\bar{l}}$

<u>Curvature</u> $R_i{}^j{}_{k\bar{l}}$ defined by

$$(\nabla_k \nabla_{\bar{l}} - \nabla_{\bar{l}} \nabla_k) \partial_i = R_i{}^j{}_{k\bar{l}} \partial_j$$

$$R_{i\bar{j}k\bar{l}} = g_{p\bar{j}} R_i{}^p{}_{k\bar{l}}$$

Note that $\nabla_k \nabla_l = \nabla_l \nabla_k$ and $\nabla_{\bar{k}} \nabla_{\bar{l}} = \nabla_{\bar{l}} \nabla_{\bar{k}}$

Ricci curvature is $R_{i\bar{j}} = g^{k\bar{l}} R_{i\bar{j}k\bar{l}}$

Scalar curvature: $R = g^{i\bar{j}} R_{i\bar{j}}$

<u>Some identities</u>

- $R_i{}^j{}_{k\bar{l}} = -\partial_{\bar{l}} \Gamma^j_{ki}$
- $R_{i\bar{j}k\bar{l}} = -\partial_k \partial_{\bar{l}} g_{i\bar{j}} + g^{p\bar{q}} (\partial_k g_{i\bar{q}})(\partial_{\bar{l}} g_{p\bar{j}})$

  $= -\partial_k \partial_{\bar{l}} g_{i\bar{j}}$ in normal coordinates

1st & 2nd Bianchi

- $R_{i\bar{j}k\bar{l}} = R_{i\bar{l}k\bar{j}} = R_{k\bar{j}i\bar{l}} = R_{k\bar{l}i\bar{j}}$
- $\nabla_p R_{i\bar{j}k\bar{l}} = \nabla_i R_{p\bar{j}k\bar{l}}$

- $(\nabla_k \nabla_{\bar{l}} - \nabla_{\bar{l}} \nabla_k) \nabla_{\bar{i}} = - R^{\bar{j}}{}_{\bar{i} k \bar{l}} \partial_{\bar{j}}$

- $R_{i\bar{j}} = - \partial_i \partial_{\bar{j}} \log \det(g_{p\bar{q}})$

<u>Definition</u> Ricci form $Ric(\omega)$ is defined

by $$Ric(\omega) = \sqrt{-1}\, R_{i\bar{j}}\, dz^i \wedge d\bar{z}^j$$

$Ric(\omega)$ is a closed real $(1,1)$-form

<u>Note</u>: if $h$ is another Kähler metric

then $\dfrac{\det(h_{p\bar{q}})}{\det(g_{p\bar{q}})}$ is independent of the

choice of coordinates $\Longrightarrow \exists f : M \to \mathbb{R}$

such that $\dfrac{\det(h_{p\bar{q}})}{\det(g_{p\bar{q}})} = f$ in any chart.

Then $$Ric(h) - Ric(g) = -\sqrt{-1}\, \partial\bar{\partial} \log f$$

so cohomology class $[Ric(g)]$ is independent

of the metric.

<u>Definition</u> 1st Chern class $c_1(M) = \dfrac{1}{2\pi} [Ric(g)] \in H^2(M, \mathbb{R})$

E.g. $\mathrm{Ric}(\omega_{FS}) = (n+1)\,\omega_{FS}$ on $\mathbb{CP}^n$

<u>Theorem</u> (Yau) Let $(M, \omega)$ be compact Kähler and let $\alpha$ be a real $(1,1)$-form such that $\frac{1}{2\pi}\alpha \in c_1(M)$. Then there is a unique Kähler metric $\eta$ with $[\eta] = [\omega]$ and $\mathrm{Ric}(\eta) = \alpha$.

Reduction to PDE:

By $\partial\bar\partial$-lemma $\quad \mathrm{Ric}(\omega) = \alpha + \sqrt{-1}\,\partial\bar\partial F$

$F : M \to \mathbb{R}$.

Suppose that we can find $f : M \to \mathbb{R}$ s.t.

$$\frac{\det(\omega + \sqrt{-1}\,\partial\bar\partial f)}{\det(\omega)} = e^F \qquad (*)$$

and $\eta = \omega + \sqrt{-1}\,\partial\bar\partial f$ is positive (Kähler)

Then $\quad -\mathrm{Ric}(\eta) + \mathrm{Ric}(\omega) = \sqrt{-1}\,\partial\bar\partial F = \mathrm{Ric}(\omega) - \alpha$

so $\quad \mathrm{Ric}(\eta) = \alpha$

uniqueness follows from $\partial\bar\partial$-lemma and uniqueness of soln. of $(*)$ up to adding a constant.